Name: ___________Suvit Kumar, Hadley Siler, Lana Bassett, Preston Ramsey______________________ **Date:** __________3/27/2026_________ **Class:** ____Engineering_____________

MyDesign Portfolio

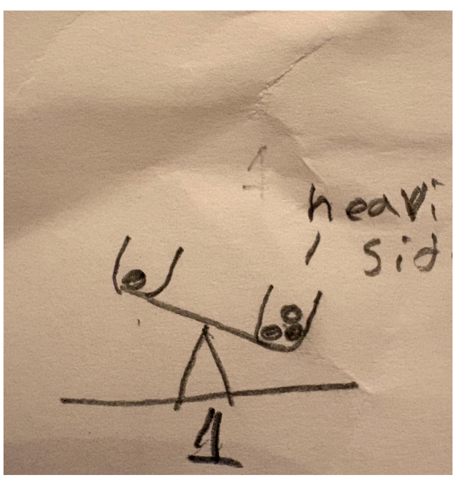
Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

Design Idea #1

Date: 3/10/26

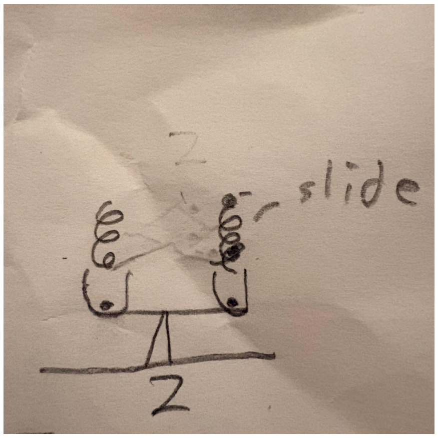
Description: ~~A balance scale that tips to show which side has more pom poms.~~



Design Idea #2

Date: 3/10/26

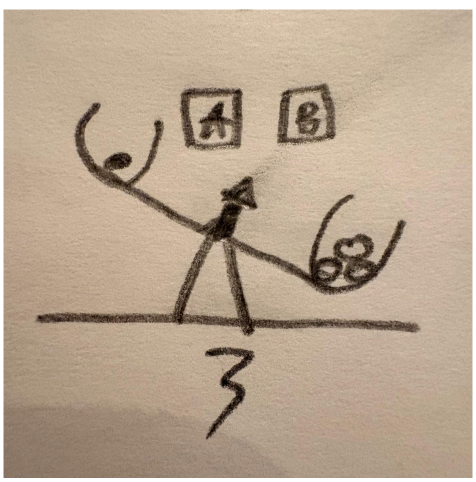
Description: Pom poms will slide into the jars down a slide similar to something found in a gumball machine.



Design Idea #3

Date: 3/10/26

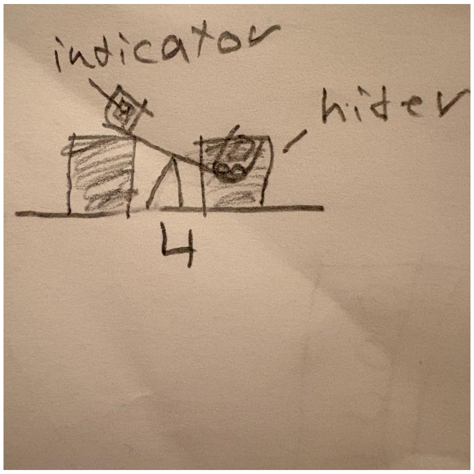
Description: The scale will have an perpendicular arrow attached to its center that will point at the sign of the winning choice.



Design Idea #4

Date: 3/10/26

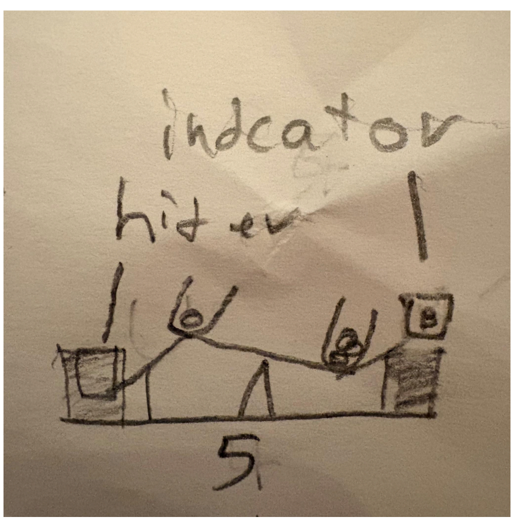
Description: An indicator card could be placed on the front of each jar on the scale containing the pom poms. As the Pom Poms fill the jar they will hide the indicator.



Design Idea #5

Date: 3/10/26

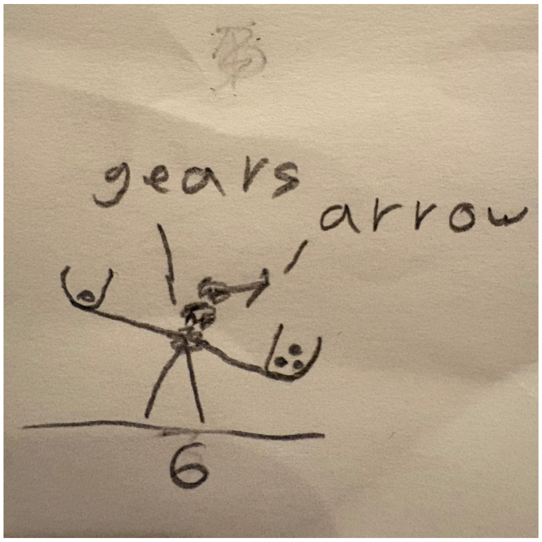
Description: As either side of the scale goes down, it lifts up an indicator to show how much each vote is winning or losing.



Design Idea #6

Date: 3/10/26

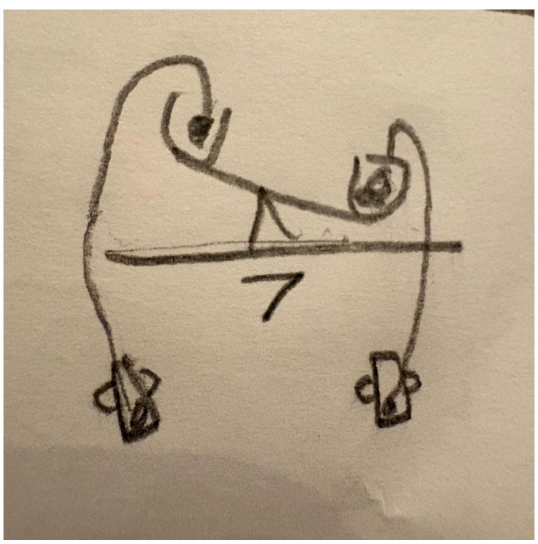
Description: Again using the weight idea we use a gear ratio to make an arrow point at the winning decision.



Design Idea #7

Date: 3/10/26

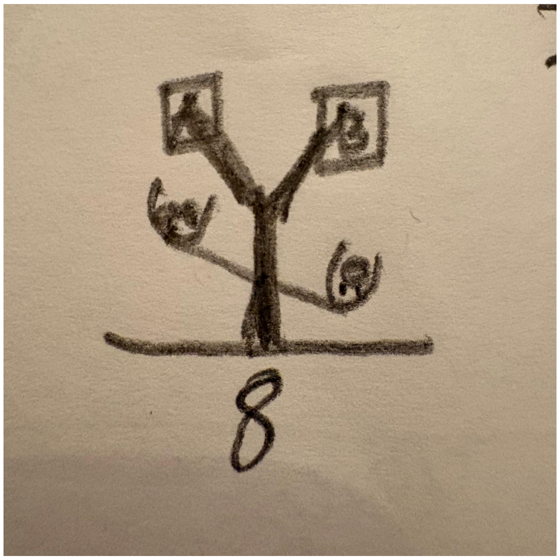
Description: Have a pom pom launcher. Two launchers each with a button where you put the pom pom into the corresponding launcher and press the button.



Design Idea #8

Date: 3/10/26

Description: A tree-like construct to hold up the signs. It'll be a pillar attached to the base of the scale and have two branches at a 120 degree angle. It's basically a glorified Y shape.



Design Idea Comparison and Selection *(D1, D2)*

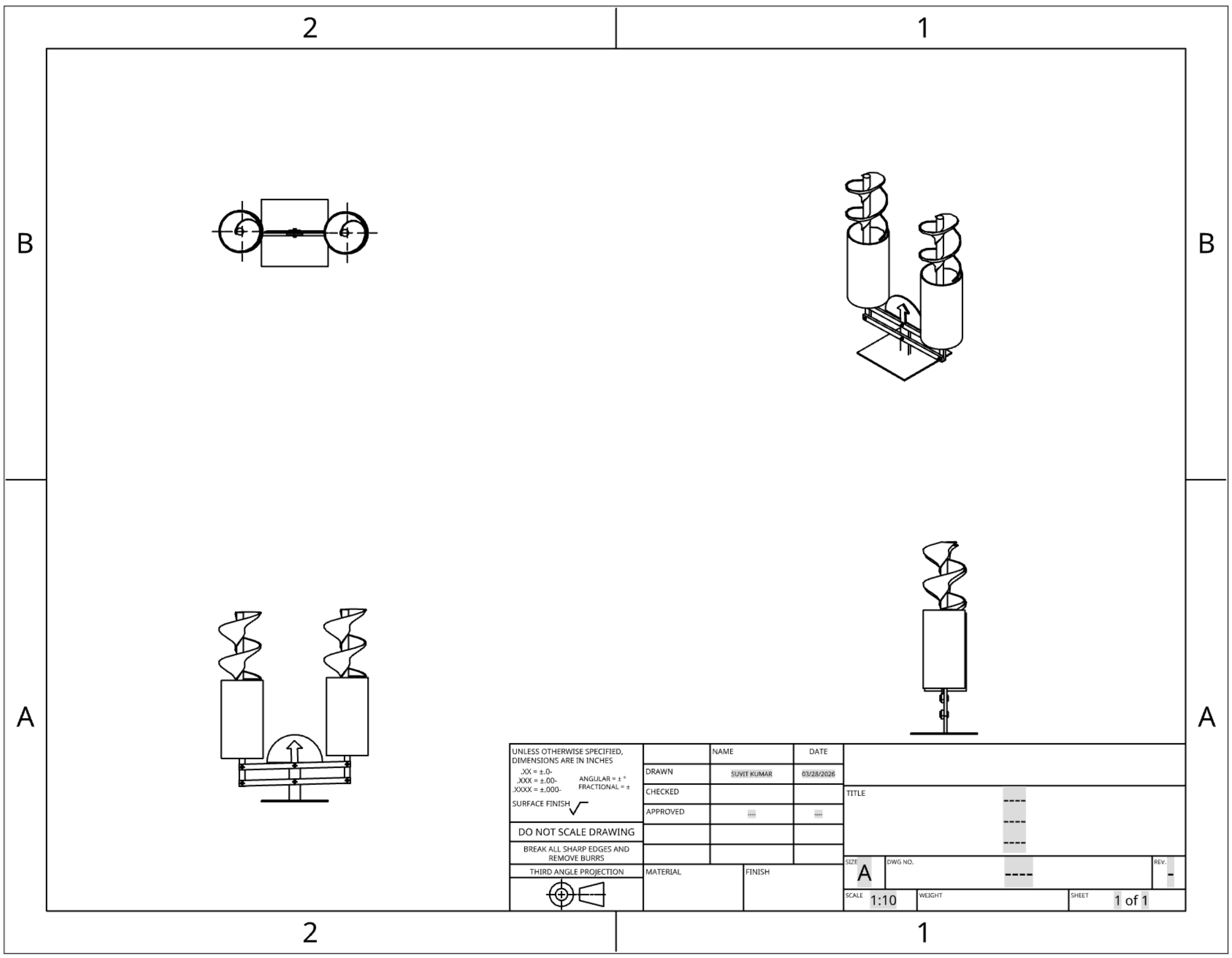
https://docs.google.com/spreadsheets/d/1CRKORJ4kxV-cmJJCQgArWm2e-_Px-_Uz/edit?gid=1026446183#gid=1026446183

We used the Element C priority rankings given and then chose different numbers to make them add up to one properly while following the priority rankings. We decided that all high priority criteria should have a weight of at least 0.1 and then we did some number balancing afterwards to make them add up to one. This is why we weighted our matrix as we did.

Design Blueprint *(D4)*

Design Blueprint #1

Date: 3/27/26
Sketch:



3D Assembly with dimensions and details (go to assembly)l: https://cad.onshape.com/documents/b54dae87b9f510f71a95c0a7/w/030c9343a50535f6bf611acb/e/e685f05062b6ead9bc7b4042

Manufacturing Methods:

Step 1: 3D Print the slide, vertical beams, horizontal beams, arrow, and base (or make it out of cardboard).

Step 2: Put the pin joints in the right holes.

Step 3: Attach the jars to the two end vertical beams with magnets.
Step 4: Attach the slide into the jar with glue.
Step 5: Create the indicator card.
Step 6: Attach the indicator card with glue.
Step 7: Attach the arrow to the top horizontal beam with glue.

Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
plastic jars	**5-10**	**2**	**10-20**
3d printer filament	**40$**	**(1)**	**40$**
Mini pompoms	**4.35$**	**1 bag has 1200pcs**	**4.35$**
Velcro	**2.99$**	**1**	**2.99$**
Total Cost			**67.34**

Plan of Action *(D3)*

March 26th: CAD
March 27th: Finalizing Construction and Testing Plan.
March 28th-31st: Making and Testing Prototype
April 1st: Miroboard Presentation.
April 7th: Document Test Results
April 9th: Making sure the stakeholders are involved, and reiterating.
April 21st: Presentation of Final Iteration